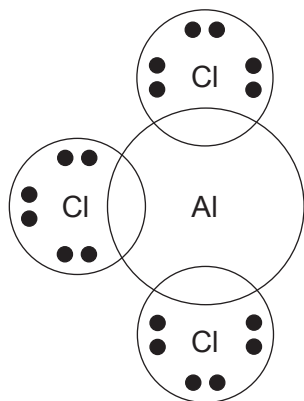


SECTION A*Answer all questions in the spaces provided.*

1. Complete the electronic structure of the Co^{2+} ion below. [1]

$1s^2 2s^2 2p^6 3s^2 3p^6$

2. The compound $\text{AlCl}_3 \cdot \text{NH}_3$ contains both covalent and coordinate bonds. [2]
Complete the dot and cross diagram for this compound.



3. Write the expression for the ionic product of water, K_w . [1]

4. The standard hydrogen electrode and hydrogen fuel cells both use the same metal as an electrode. Name the metal and explain why this metal is suitable. [1]

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